

RAI-818: INTELLIGENT **MOBILE ROBOTICS**

Probabilistic Robotics – II

- **Bayesian Filtering (Cont ...)**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:

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Bayesian Filtering

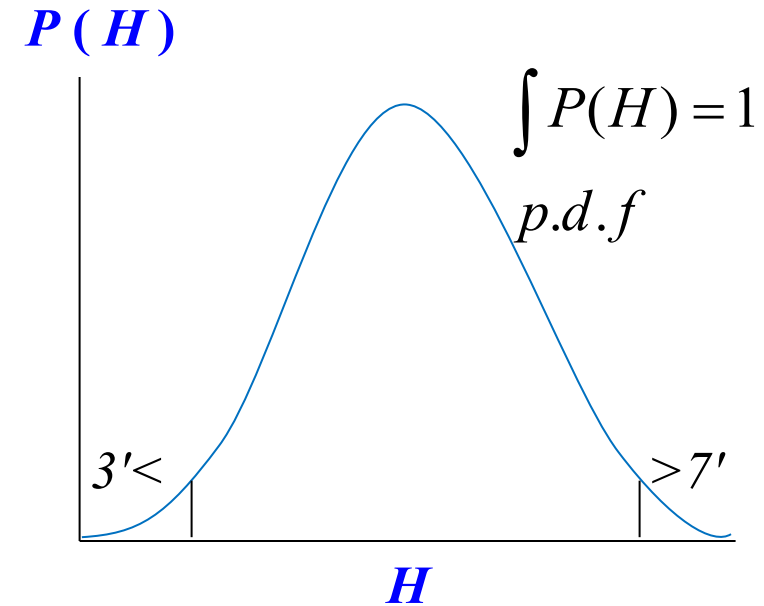


- Localization
- Probability
- **Bayesian Filtering**
 - Bayes Theorem
 - **Bayesian Inference**

Bayesian Inference

$$P(H|D) = \frac{P(H) \times P(D|H)}{P(D)}$$

- How much do you believe in all possible hypothesis?
- Say all the different possible heights
- 2 ft, 3ft, 7ft, 8ft heights are not very likely therefore at the end of the distribution
- 5-6 ft very likely therefore in the middle of the distribution



Heights of persons
State-space of possibilities



Defining futures

Bayesian Filtering

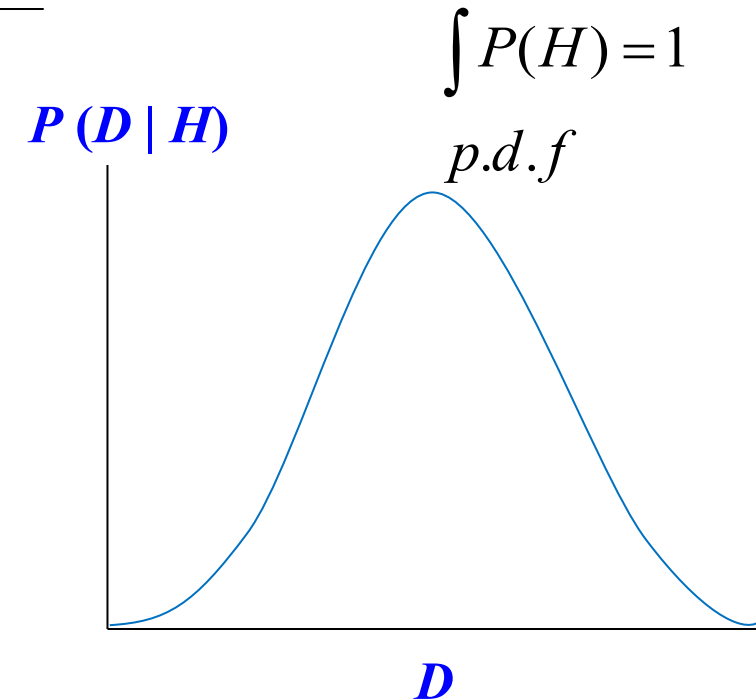


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Bayesian Inference

$$P(H|D) = \frac{P(H) \times P(D|H)}{P(D)}$$

- The same cannot be done for the likelihood $P(D|H)$
- Normally the probability $P(D|H)$ over (D) will be 1, which is not what we want.
- We are interested in the (H) which can be found by the joint distribution $P(D,H)$ which will be different



Defining futures

Bayesian Filtering

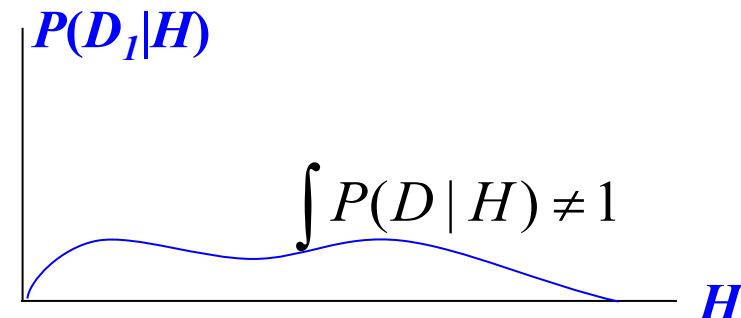
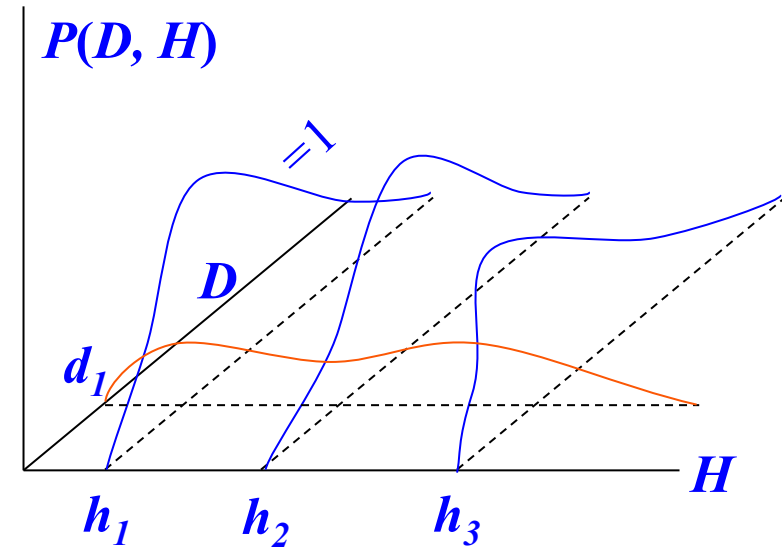


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Bayesian Inference

$$P(H|D) = \frac{P(H) \times P(D|H)}{P(D)}$$

- If we take a certain hypothesis h_1 and draw the distribution of this over the data (i.e., given a particular hypothesis h_1 what is the probability of the data)
- We want to know the prob for a specific data point d_1 over all hypothesis
- This will not be 1 and is not specifically a prob distribution and needs to be normalized which is the purpose of the denominator $P(D)$



Bayesian Filtering

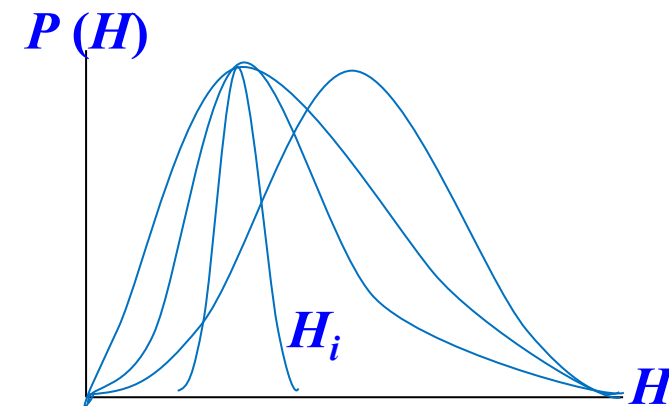
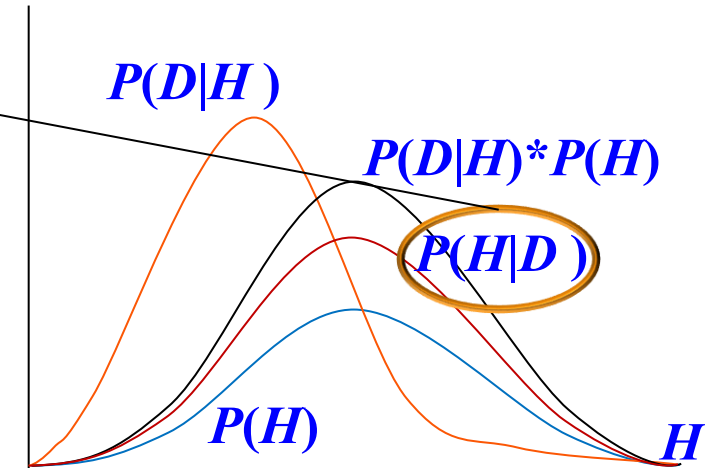


- Localization
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Bayesian Inference

$$P(H|D) = \frac{P(H) \times P(D_2|H)}{P(D)}$$

- If we look at the numerator it is just the multiplication of the prob distribution of the prior and likelihood
- To normalize $P(D|H)$ so that it can be multiplied with $P(H)$ we divide by $P(D)$ (which is normally difficult to determine)
- What we can typically do is find the area of the curve $P(D|H)*P(H)$ and divide each point by that value which will give us $P(H|D)$ which will have a $p.d.f = 1$



Defining futures

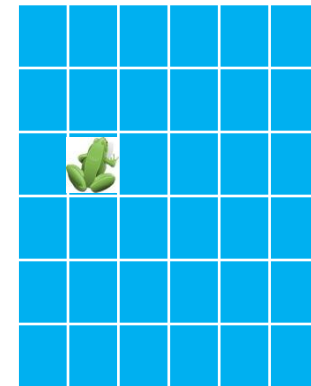
Bayesian Filtering



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 - **Example 1**

Bayesian Example

- One of Dr Faraz's researchers/students from this class has been asked to capture a frog from the pond so that its motion can be modeled
- Suppose the frog is hidden in the pond and is truly at the location shown
- However, the researcher knows that since it is night, therefore, when he hears the sound the frog is may not actually be where he thinks it is (i.e., there is some noise/error)
- The researcher does not want to waste time/energy doing a brute force search (since Dr Faraz will not like it) so he will only attack the frog when he is very sure about its location.
- To reduce the effect of noise the researcher decides to iteratively update his estimate regarding the location of the frog



Defining futures

Bayesian Filtering

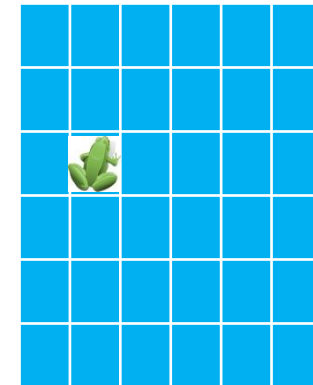
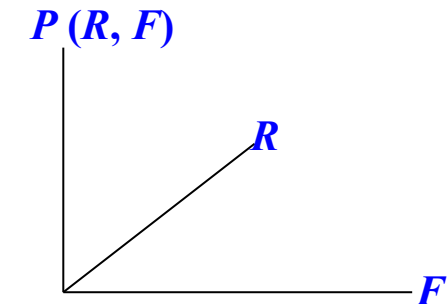
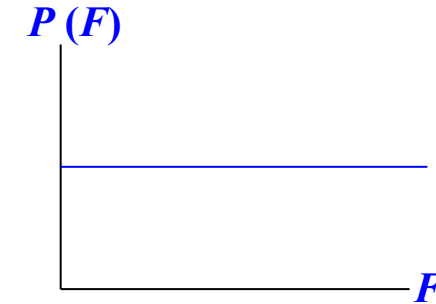


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Bayesian Example

$$P(F|R) = \frac{P(F) \times P(R|F)}{P(R)}$$

- Lets look at it from a Bayesian point of view
- Lets say at the start the researcher does not have a clue about the location of the frog. This can be represented as a uniform distribution
- For the likelihood we need to get the joint distribution of $P(R, F)$ which is hard to get empirically or estimate since there are numerous positions
- Therefore, we rely on some kind of expectation about the world to constrain the likely location of the frog.



Defining futures

Bayesian Filtering

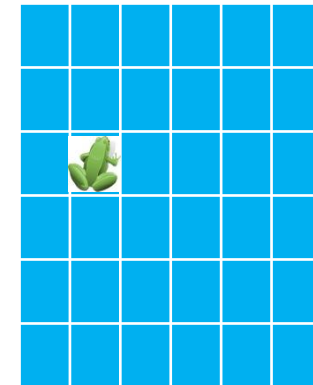
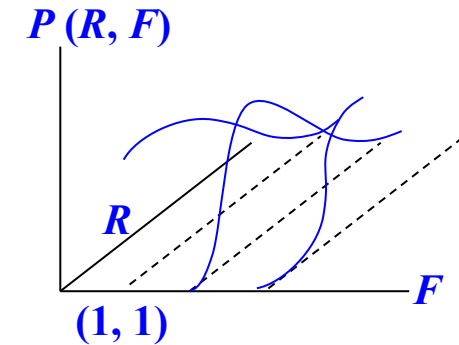
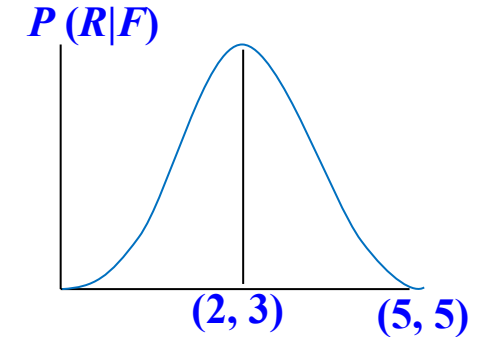


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Bayesian Example

$$P(F|R) = \frac{P(F) \times P(R|F)}{P(R)}$$

- Lets say that the frog can only make a sound in such a way that it gives away its position with a Gaussian distribution.
- Therefore, our 3D plot will have different Gaussian distributions for the sound based on where the frog is actually located as shown
- If we assume a Gaussian distribution we can build our function quite easily
- We assume for every state F the Gaussian distribution of the sound R



Defining futures

Bayesian Filtering

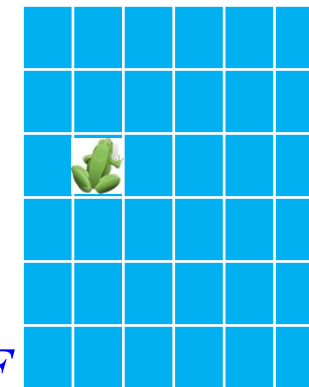
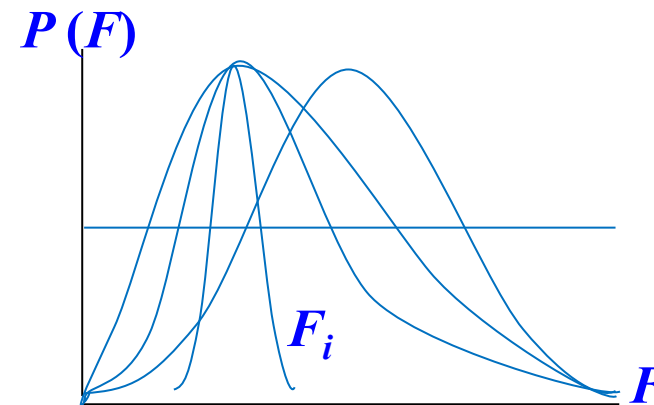
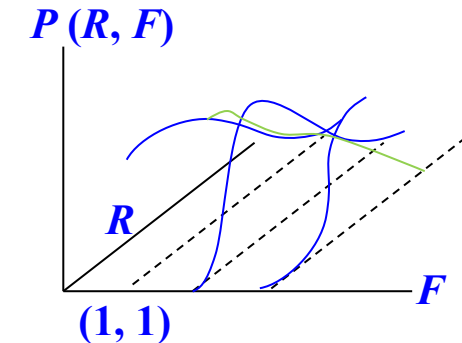
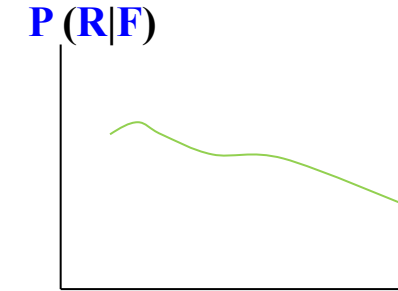


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Bayesian Example

$$P(F|R) = \frac{P(F) \times P(R|F)}{P(R)}$$

- For a particular R we can get a particular curve which will give us $P(R|F)$
- We can combine the likelihood data with our prior expectations to iteratively estimate with each new sound where we think the frog is located.
- When the researcher is confident he will attack and capture the frog



Defining futures

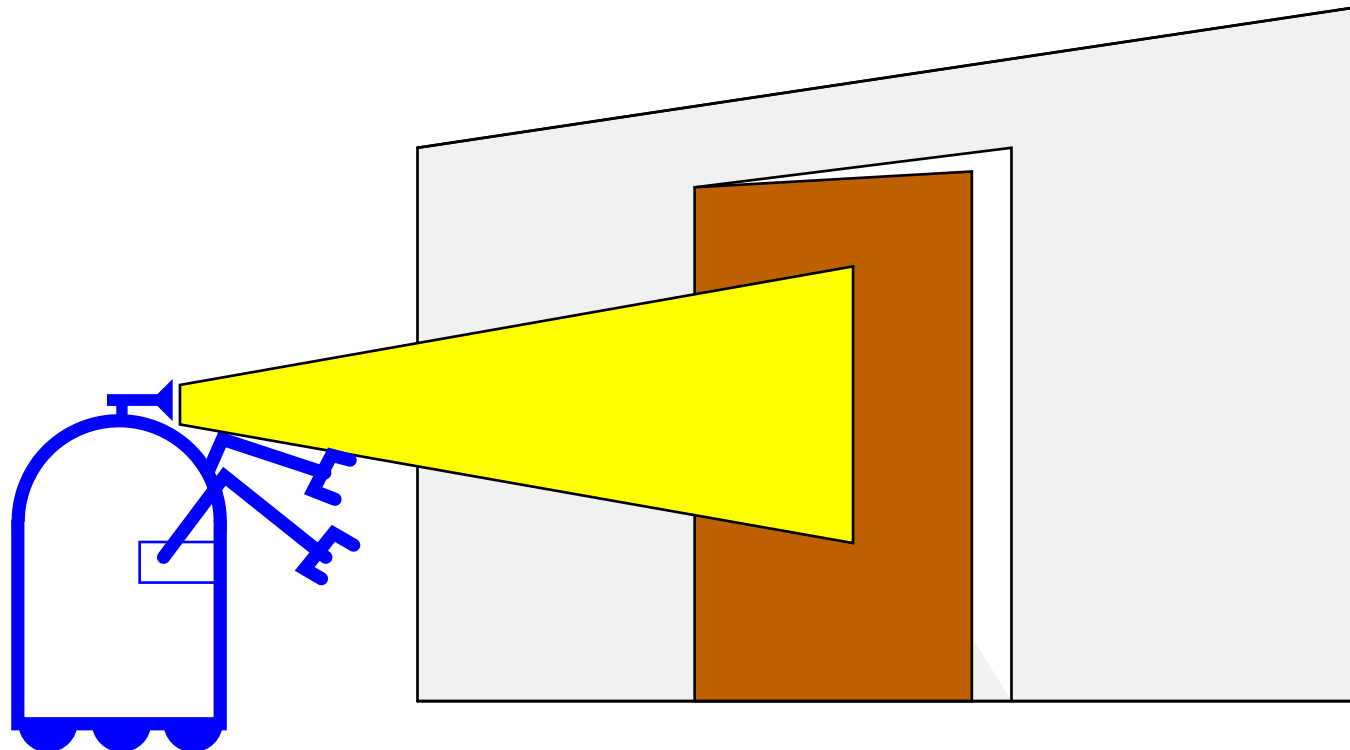
Bayesian Filtering



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 - **Example 2**
 - State Estimation

State Estimation

- A door can be in two possible states, i.e., *open* or *close*
- A robot estimates the state of a door using its camera
- Only the robot can change the state of the door, using the action *push* to a open door



Defining futures

Bayesian Filtering

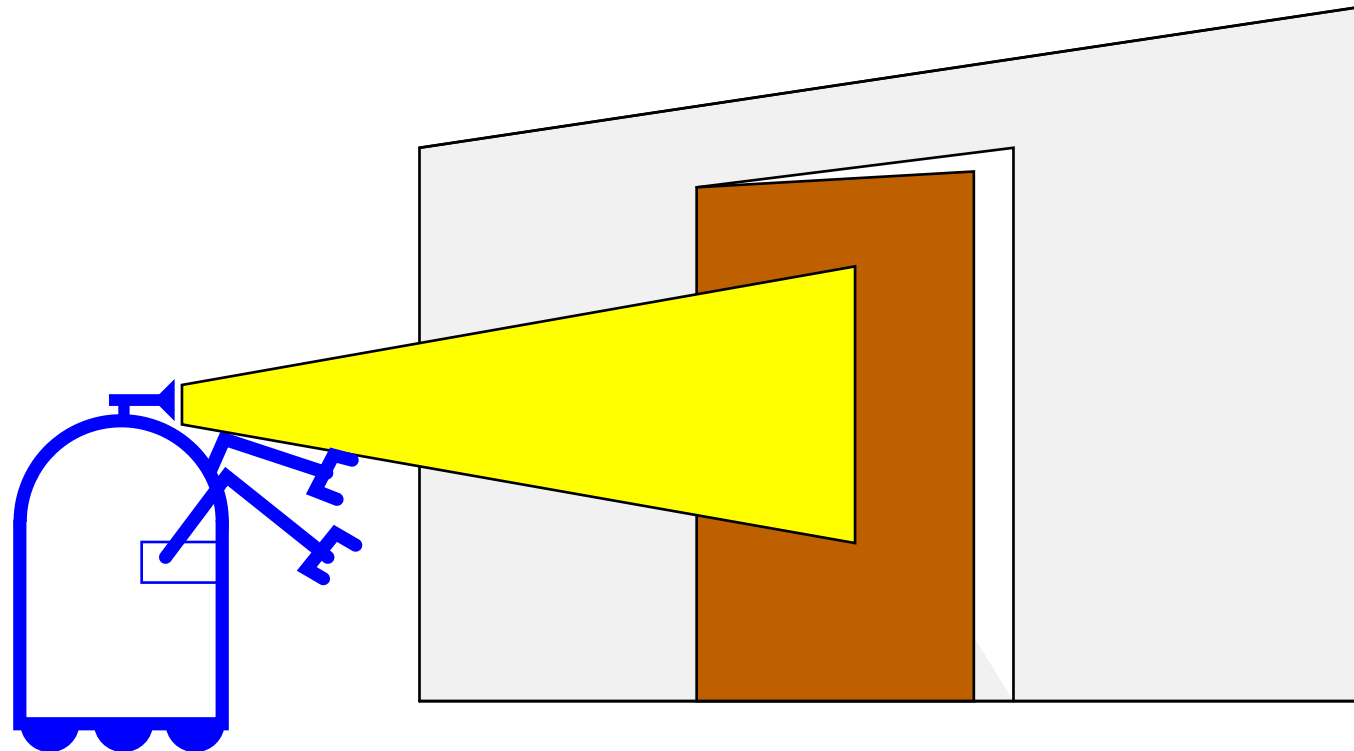


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State Estimation

- Suppose a robot obtains measurement z
- What is $P(X_t = \text{open} \mid Z_t = \text{sense-open})$?

$$\text{bel}(X_0 = \text{open}) = \text{bel}(X_0 = \neg \text{open}) = 0.5$$



Defining futures



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 - Causal Vs Diagnostic Reasoning

Causal vs Diagnostic Reasoning

- $P(X_t = open \mid Z_t = sense-open)$ is diagnostic.
- $P(Z_t = sense-open \mid X_t = open)$ is causal.
- Often **causal** knowledge is easier to obtain.
- Bayes rule allows us to use causal knowledge:

$$P(open|z) = \frac{P(z|open)P(open)}{P(z)}$$



Defining futures



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Causal vs Diagnostic Reasoning - Example

- $P(z|open) = 0.6$ $P(z|\neg open) = 0.3$
- $P(open) = P(\neg open) = 0.5$

$$P(open | z) = \frac{P(z | open)P(open)}{P(z | open)p(open) + P(z | \neg open)p(\neg open)}$$

$$P(open | z) = \frac{0.6 \cdot 0.5}{0.6 \cdot 0.5 + 0.3 \cdot 0.5} = \frac{2}{3} = 0.67$$

- **z raises the probability that the door is open.**



Defining futures

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 - Combining Evidence

Combining Evidence

- Suppose our robot obtains another observation z_2 .
- How can we integrate this new information?
- More generally, how can we estimate $P(x | z_1 \dots z_n)$?



Defining futures

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 - **Recursive Updating**

Recursive Bayesian Updating

$$P(x \mid z_1, \dots, z_n) = \frac{P(z_n \mid x, z_1, \dots, z_{n-1}) P(x \mid z_1, \dots, z_{n-1})}{P(z_n \mid z_1, \dots, z_{n-1})}$$

Markov assumption: z_n is **independent** of $z_1, \dots, z_{n-1}$ if we know x .

$$\begin{aligned} P(x \mid z_1, \dots, z_n) &= \frac{P(z_n \mid x) P(x \mid z_1, \dots, z_{n-1})}{P(z_n \mid z_1, \dots, z_{n-1})} \\ &= \eta P(z_n \mid x) P(x \mid z_1, \dots, z_{n-1}) \\ &= \eta_{1 \dots n} \prod_{i=1 \dots n} P(z_i \mid x) P(x) \end{aligned}$$



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 - **Additional Measurements**

2nd Measurement Example

- $P(z_2 | open) = 0.5$
- $P(z_2 | \neg open) = 0.6$
- $P(open | z_1) = 2/3$

$$\begin{aligned} P(open | z_2, z_1) &= \frac{P(z_2 | open) P(open | z_1)}{P(z_2 | open) P(open | z_1) + P(z_2 | \neg open) P(\neg open | z_1)} \\ &= \frac{\frac{1}{2} \cdot \frac{2}{3}}{\frac{1}{2} \cdot \frac{2}{3} + \frac{3}{5} \cdot \frac{1}{3}} = \frac{5}{8} = 0.625 \end{aligned}$$

- z_2 lowers the probability that the door is open.



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 - **Actions**

Actions

- Often the world is **dynamic** since
 - actions carried out by the robot,
 - actions carried out by other agents,
 - or just the **time** passing by change the world.
- How can we **incorporate** such **actions**?



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Typical Actions

Action at time t is denoted u .

- The robot turns its wheels to move
- The robot uses its manipulator to grasp an object
- Plants grow over time...

Note that

- Actions are never carried out with absolute certainty.
- In contrast to measurements, actions generally increase the uncertainty.



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Modelling Actions

- The outcome of an action u is modelled by the conditional probability distribution
$$P(x \mid u, x')$$
- That is, the probability that, when in state x' , executing action u , changes the state to x .



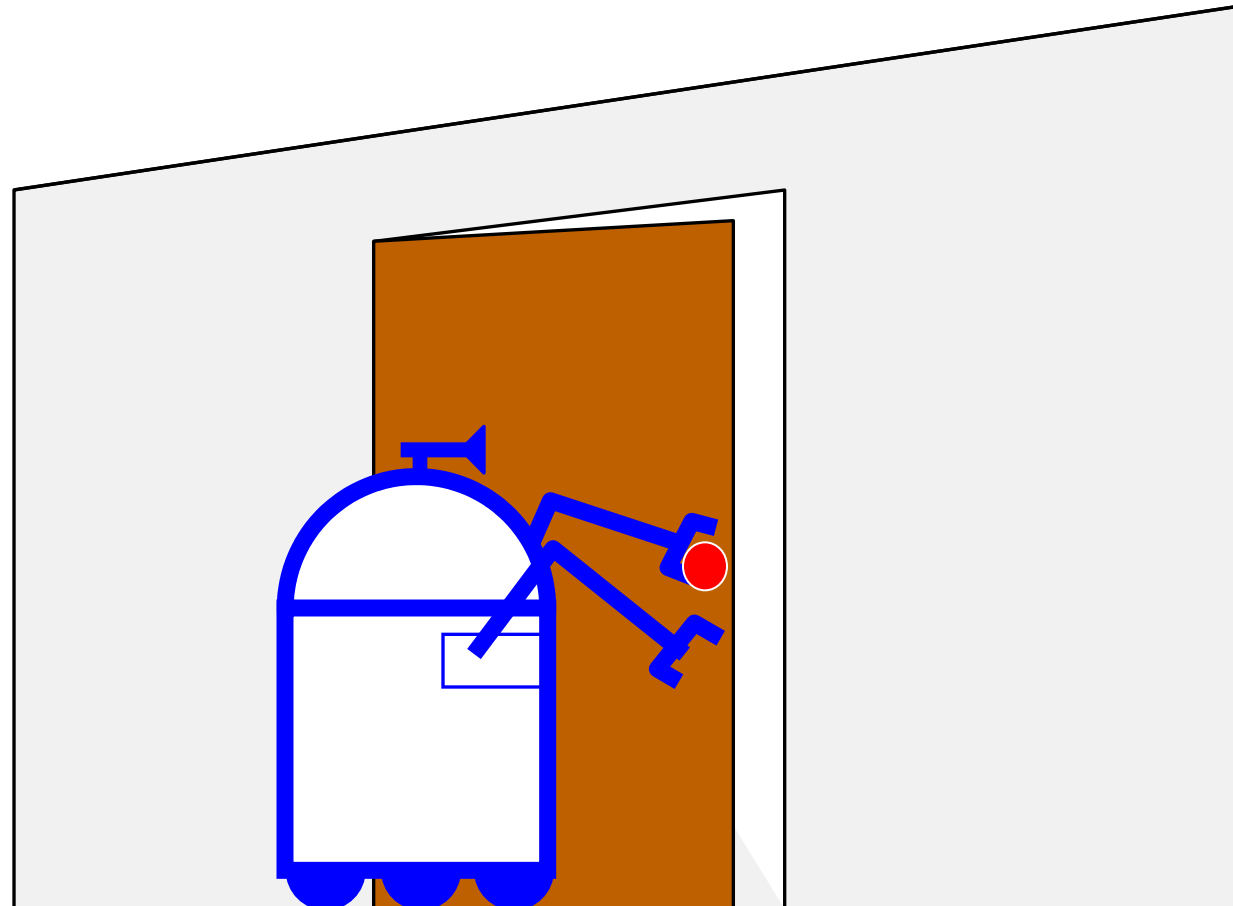
Defining futures

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Example Door Closing



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Example Door Closing

- We assume that the robot *does not know* the state of the door initially

$$bel(X_0 = open) = 0.5$$

$$bel(X_0 = \neg open) = 0.5$$

- Further we assume that the robot's sensors are noisy

$$P(Z_t = sense-open \mid X_t = open) = 0.6$$

$$P(Z_t = sense-closed \mid X_t = open) = 0.4$$

$$P(Z_t = sense-open \mid X_t = closed) = 0.2$$

$$P(Z_t = sense-closed \mid X_t = closed) = 0.8$$



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Example Door Closing

- Assume that if an *agent tries to open the door* there is a possibility of failure

$$P(X_t = \text{open} \mid U_t = \text{push}, X_{t-1} = \text{open}) = 1$$

$$P(X_t = \text{closed} \mid U_t = \text{push}, X_{t-1} = \text{open}) = 0$$

$$P(X_t = \text{open} \mid U_t = \text{push}, X_{t-1} = \text{closed}) = 0.8$$

$$P(X_t = \text{closed} \mid U_t = \text{push}, X_{t-1} = \text{closed}) = 0.2$$

- Finally, the *agent can choose to do nothing* in which case the world does not change

$$P(X_t = \text{open} \mid U_t = \text{do-nothing}, X_{t-1} = \text{open}) = 1$$

$$P(X_t = \text{closed} \mid U_t = \text{do-nothing}, X_{t-1} = \text{open}) = 0$$

$$P(X_t = \text{open} \mid U_t = \text{do-nothing}, X_{t-1} = \text{closed}) = 0$$

$$P(X_t = \text{closed} \mid U_t = \text{do-nothing}, X_{t-1} = \text{closed}) = 1$$



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Example Door Closing

Bayes- Filter ($bel(X_0), U_t = do-nothing, Z_t = sense-open$)

Prediction Step: $\overline{bel}(X_1) = \sum_{X_0} P(X_1 | U_1, X_0) bel(X_0)$

So,

$$\begin{aligned}\overline{bel}(X_1 = open) &= P(X_1 = open | U_t = do-nothing, X_0 = open) bel(X_0 = open) \\ &\quad + P(X_1 = open | U_t = do-nothing, X_0 = closed) bel(X_0 = closed) \\ &= (1 * 0.5 + 0 * 0.5) = 0.5\end{aligned}$$

Similarly,

$$\begin{aligned}\overline{bel}(X_1 = closed) &= P(X_1 = closed | U_t = do-nothing, X_0 = open) bel(X_0 = open) \\ &\quad + P(X_1 = closed | U_t = do-nothing, X_0 = closed) bel(X_0 = closed) \\ &= (0 * 0.5 + 1 * 0.5 + 0) = 0.5\end{aligned}$$



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Example Door Closing

Measurement Update

$$bel(X_1) = \eta P(Z_t = \textit{sense-open} \mid X_1) \overline{bel}(X_1)$$

So,

$$\begin{aligned} bel(X_1 = \textit{open}) &= \eta P(Z_1 = \textit{sense-open} \mid X_1 = \textit{open}) \overline{bel}(X_1 = \textit{open}) \\ &= \eta 0.6 * 0.5 = \eta 0.3 \end{aligned}$$

Similarly,

$$\begin{aligned} bel(X_1 = \textit{closed}) &= \eta P(Z_1 = \textit{sense-open} \mid X_1 = \textit{closed}) \overline{bel}(X_1 = \textit{closed}) \\ &= \eta 0.2 * 0.5 = \eta 0.1 \end{aligned}$$

The normalization constant can be calculated as $\eta = (0.3 + 0.1)^{-1} = 2.5$

Finally, we have

$$bel(X_1 = \textit{open}) = 0.75$$

$$bel(X_1 = \textit{closed}) = 0.25$$

These values can be iterated for the next time step



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Bayes- Filter ($bel(X_1) = 0.5, U_t = push, Z_t = sense-open$)

Prediction Step: $\overline{bel}(X_2) = \sum_{X_1} P(X_2 | U_2, X_1) bel(X_1)$

So,

$$\begin{aligned}\overline{bel}(X_2 = open) &= P(X_2 = open | U_t = push, X_1 = open) bel(X_1 = open) \\ &\quad + P(X_2 = open | U_t = push, X_1 = closed) bel(X_1 = closed) \\ &= (1 * 0.75 + 0.8 * 0.25) = 0.95\end{aligned}$$

Similarly,

$$\begin{aligned}\overline{bel}(X_2 = closed) &= P(X_2 = closed | U_t = do-nothing, X_1 = open) bel(X_1 = open) \\ &\quad + P(X_2 = closed | U_t = do-nothing, X_1 = closed) bel(X_1 = closed) \\ &= (0 * 0.75 + 0.2 * 0.25) = 0.05\end{aligned}$$



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Measurement Update

$$bel(X_2) = \eta P(Z_t = \text{sense-open} \mid X_2) \overline{bel}(X_2)$$

So,

$$\begin{aligned} bel(X_2 = \text{open}) &= \eta P(Z_2 = \text{sense-open} \mid X_2 = \text{open}) \overline{bel}(X_2 = \text{open}) \\ &= \eta 0.6 * 0.95 = 0.98 \end{aligned}$$

Similarly,

$$\begin{aligned} bel(X_2 = \text{closed}) &= \eta P(Z_2 = \text{sense-open} \mid X_2 = \text{closed}) \overline{bel}(X_2 = \text{closed}) \\ &= \eta 0.2 * 0.05 = 0.017 \end{aligned}$$

The normalization constant can be calculated as $\eta = (0.57 + 0.01)^{-1} = 1.724$



Defining futures